

✓ IDC 222 – Practice Questions (STL Set & Map)

1. Find and Remove Elements in Range

Write a C++ program using `set` container that:

- Read n integers into a set
- Takes two values `low` and `high` as input
- Displays all elements in the range `[low, high]`
- Removes all elements in this range from the set
- Displays the final set after removal

Note: Do not use `lower_bound()` or `upper_bound()` functions.

Example:

Input:

```
Set size: 10
Elements: 5 10 15 20 25 30 35 40 45 50
Enter range to remove:
Low: 20
High: 40
```

Output:

```
Original set: 5 10 15 20 25 30 35 40 45 50
Elements in range [20, 40]: 20 25 30 35 40
Removing elements in range...
Set after removal: 5 10 15 45 50
```

2. Symmetric Difference using Set

Write a C++ program using `set` container to find symmetric difference:

- Read n elements for Set A and m elements for Set B.
- Display Set A and Set B in sorted order
- Find and display elements only in Set A, only in Set B, and common elements in both sets.
- Find and display the symmetric difference (elements in A or B but not in both).

Symmetric Difference = $(A - B) \cup (B - A)$

Example:

Input:

```
Enter number of elements in Set A: 5
Enter elements of Set A:
1 2 3 4 5
Enter number of elements in Set B: 4
Enter elements of Set B:
3 4 6 7
```

Output:

```
Set A: 1 2 3 4 5
Set B: 3 4 6 7

Elements only in A: 1 2 5
Elements only in B: 6 7
Common elements: 3 4
Symmetric Difference: 1 2 5 6 7
```

3. Subset Check using Set

Write a C++ program using the set container to:

- Read n characters for Set A
- Read m characters for Set B
- Display both sets in sorted order
- Check and display:
 - Whether Set B is a subset of Set A
 - Whether Set A is a subset of Set B
 - Whether Set A and Set B are equal

Example:

Input:

```
Enter number of elements in Set A: 5
Enter characters of Set A:
a c e b d

Enter number of elements in Set B: 3
Enter characters of Set B:
a c e
```

Output:

```
Set A: a b c d e
```

```
Set B: a c e
```

```
Set B is a subset of Set A
```

```
Set A is NOT a subset of Set B
```

```
Set A and Set B are NOT equal
```

4. Country Code Directory using Map

Write a C++ program using the map container to create a country code directory.

The program should:

- Insert 3 country details (country code as **key**, country name as **value**)
- Display all countries with their codes in sorted order using `iterator`
- Search for a country using its country code
- Delete a country using its country code
- Display all countries after deletion using `range-based for loop`

Example:

Input:

```
Enter 3 country details:
```

```
Country 1 - Code: 91
```

```
Country 1 - Name: India
```

```
Country 2 - Code: 1
```

```
Country 2 - Name: USA
```

```
Country 3 - Code: 44
```

```
Country 3 - Name: UK
```

```
Enter country code to search: 91
```

```
Enter country code to delete: 44
```

Output:

```
--- All Countries ---
Code: 1, Name: USA
Code: 44, Name: UK
Code: 91, Name: India

--- Search Result ---
Country code 91 found: India

--- After Deletion ---
Country code 44 deleted successfully!

--- All Countries After Deletion ---
Code: 1, Name: USA
Code: 91, Name: India
```

5. Blood Group Donor Registry (Menu-Driven)

Write a C++ menu-driven program using the `map` container to manage blood donors where:

1. **Key:** Blood Group (char) - 'A', 'B', 'O'
2. **Value:** Donor name (string)

Menu Options:

- Register a donor
- Find donor for a blood group
- Update donor for a blood group
- Remove a donor
- Display all blood groups and donors
- Exit

Example: Input/Output

```
----- MENU -----
1. Register a donor
2. Find donor for a blood group
3. Update donor for a blood group
4. Remove a donor
5. Display all blood groups and donors
6. Exit
-----
```

Enter your choice: 1

Enter blood group (A, B, O): A

Enter donor name: John Doe

Donor registered successfully!

Enter your choice: 1

Enter blood group (A, B, O): B

Enter donor name: Jane Smith

Donor registered successfully!

Enter your choice: 1

Enter blood group (A, B, O): O

Enter donor name: Mike Johnson

Donor registered successfully!

Enter your choice: 5

--- All Blood Groups and Donors ---

Blood Group: A, Donor: John Doe

Blood Group: B, Donor: Jane Smith

Blood Group: O, Donor: Mike Johnson

Total blood groups registered: 3

Enter your choice: 2

Enter blood group to search: B

Donor found!

Blood Group: B, Donor: Jane Smith

Enter your choice: 1

Enter blood group (A, B, O): B

Enter donor name: Robert Brown

Blood group 'B' already has a donor !

Current donor: Jane Smith

Use option 3 to update donor.

Enter your choice: 3

Enter blood group to update: A

Enter new donor name: Alice Green

Donor updated successfully!

Enter your choice: 4

Enter blood group to remove: O

Donor removed successfully!

Enter your choice: 5

--- All Blood Groups and Donors ---

Blood Group: A, Donor: Alice Green

Blood Group: B, Donor: Jane Smith

Total blood groups registered: 2

Enter your choice: 6

Exiting... Thank you!